

Numerical Simulation of Financial Stochastic Differential Equations

From Euler–Maruyama to Nested-Grid Verification

Group Members

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Numerical Partial Differential Equations

September 2026

Purpose of the Presentation

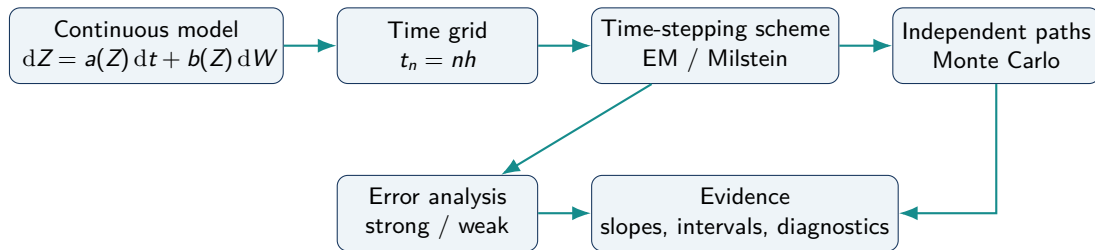
Intended audience

Students familiar with elementary probability and the explicit Euler method, but with no prior background in stochastic calculus or mathematical finance.

By the end of the presentation, the audience should be able to explain:

- ① how an SDE is discretised on a time grid;
- ② why Monte Carlo averages estimate expectations;
- ③ how Euler–Maruyama and Milstein differ;
- ④ what strong and weak convergence measure;
- ⑤ how to validate a solver when no exact solution is available;
- ⑥ what the numerical evidence does and does not establish.

The Numerical Workflow



Central numerical-analysis question

A simulation is credible only when the discretisation, error reduction, statistical uncertainty, and reproducibility are all made explicit.

Connection with Numerical PDEs

For the diffusion

$$dS_t = a(S_t) dt + b(S_t) dW_t,$$

define

$$u(t, s) = \mathbb{E}[\varphi(S_T) \mid S_t = s].$$

Under standard assumptions, u satisfies the backward Kolmogorov equation

$$u_t + a(s)u_s + \frac{1}{2}b(s)^2 u_{ss} = 0, \quad u(T, s) = \varphi(s).$$

Two numerical routes

- ▶ discretise the PDE in (t, s) ;
- ▶ discretise sample paths in time and approximate the expectation by Monte Carlo.

This coursework uses the second route while retaining the central themes of **discretisation**, **convergence**, **stable computation**, and **error quantification**.

Two Models, Two Numerical Roles

	Model 1: GBM benchmark	Model 2: non-affine coupled model
Purpose	exact solution for verification	realistic test without a closed form
Methods	exact update, EM, Milstein	log-variable Euler–Maruyama
Main outputs	strong and weak errors, moments	nested-grid differences and nonlinear observables
Reference	exact terminal value on the same path	a much finer numerical grid
Main difficulty	separating pathwise and expectation errors	controlling reference-solution error

The exact benchmark validates the implementation before it is used in a model without an exact solution.

Structure

1. From the Euler Method to Stochastic Time Stepping
2. Model 1: Verification against an Exact Solution
3. Model 2: Nested-Grid Verification without an Exact Solution
4. Evaluation, Reproducibility, and Conclusions

1. From the Euler Method to Stochastic Time Stepping

Review: Explicit Euler for an ODE

Consider

$$\frac{dy}{dt} = f(y), \quad y(0) = y_0,$$

on the grid

$$0 = t_0 < t_1 < \cdots < t_N = T, \quad h = \frac{T}{N}.$$

Freezing the slope over one time step gives

$$y(t_{n+1}) - y(t_n) \approx f(y_n)h, \quad \boxed{y_{n+1} = y_n + f(y_n)h}.$$

Key idea

The continuous evolution is replaced by many short steps. A smaller h usually improves the approximation, at greater computational cost.

An SDE Adds a Random Increment

A scalar stochastic differential equation is

$$\mathrm{d}Z_t = a(Z_t) \mathrm{d}t + b(Z_t) \mathrm{d}W_t.$$

- ▶ $a(Z_t) \mathrm{d}t$: drift, analogous to deterministic ODE motion;
- ▶ $b(Z_t) \mathrm{d}W_t$: diffusion, representing random fluctuations;
- ▶ W_t : Brownian motion, whose paths are not differentiable.

Over one interval,

$$Z_{t_{n+1}} - Z_{t_n} = \int_{t_n}^{t_{n+1}} a(Z_t) \mathrm{d}t + \int_{t_n}^{t_{n+1}} b(Z_t) \mathrm{d}W_t.$$

Freezing both coefficients at the left endpoint produces the stochastic Euler method.

Brownian Increments in the Code

Brownian increments satisfy

$$\Delta W_n = W_{t_{n+1}} - W_{t_n} \sim \mathcal{N}(0, h), \quad \Delta W_n = \sqrt{h} Z_n, \quad Z_n \sim \mathcal{N}(0, 1).$$

Mean

$$\mathbb{E}[\Delta W_n] = 0.$$

Positive and negative disturbances cancel on average.

Scale

$$\text{SD}(\Delta W_n) = \sqrt{h}.$$

The random increment is $O(\sqrt{h})$ rather than $O(h)$.

Common implementation error

Using hZ_n suppresses the stochastic term too rapidly and changes the limiting model.

Freeze both coefficients during one step:

$$\int_{t_n}^{t_{n+1}} a(Z_t) dt \approx a(Z_n)h, \quad \int_{t_n}^{t_{n+1}} b(Z_t) dW_t \approx b(Z_n)\Delta W_n.$$

The Euler–Maruyama (EM) method is

$$Z_{n+1} = Z_n + a(Z_n)h + b(Z_n)\Delta W_n.$$

- ▶ It is explicit: only the known state Z_n is used.
- ▶ One sequence of increments generates one approximate path.
- ▶ Many paths are required to estimate expectations and uncertainty.

Monte Carlo Estimation

To estimate $\mathbb{E}[X]$, generate independent samples $X^{(1)}, \dots, X^{(M)}$:

$$\hat{m}_M = \frac{1}{M} \sum_{i=1}^M X^{(i)}.$$

The law of large numbers gives $\hat{m}_M \rightarrow \mathbb{E}[X]$, while

$$\text{SE}(\hat{m}_M) \approx \frac{s_X}{\sqrt{M}}.$$

An approximate 95% confidence interval is

$$\boxed{\hat{m}_M \pm 1.96 \frac{s_X}{\sqrt{M}}}.$$

Cost implication

Halving the Monte Carlo standard error requires approximately four times as many paths.

Two Contributions to the Total Error

$$\underbrace{\hat{Q}_{h,M} - Q}_{\text{total error}} = \underbrace{(\mathbb{E}[Q_h] - Q)}_{\text{time-discretisation bias}} + \underbrace{(\hat{Q}_{h,M} - \mathbb{E}[Q_h])}_{\text{sampling noise}}.$$

Typical magnitudes are

$$|\text{bias}| \approx Ch^p, \quad |\text{sampling error}| \approx C_M M^{-1/2}.$$

Experimental design principle

To measure a time-convergence rate, sampling noise must remain below the discretisation error over the fitted grid range.

Why Stochastic Taylor Expansions Need a Correction

For smooth f , Itô's formula is

$$df(Z_t) = f'(Z_t) dZ_t + \frac{1}{2} f''(Z_t) b(Z_t)^2 dt.$$

The final term is absent from the ordinary chain rule. It appears because

$$\Delta W = O(\sqrt{h}) \implies (\Delta W)^2 = O(h).$$

Sufficient intuition for this coursework

Some second-order random terms are as large as the deterministic drift term and therefore cannot be discarded.

Milstein: One Additional Random Second-Order Term

For $dZ = a(Z)dt + b(Z)dW$, the scalar Milstein method is

$$Z_{n+1} = Z_n + a(Z_n)h + b(Z_n)\Delta W_n + \frac{1}{2}b(Z_n)b'(Z_n)((\Delta W_n)^2 - h).$$

- ▶ The first three terms are exactly Euler–Maruyama.
- ▶ The correction has zero mean because $\mathbb{E}[(\Delta W)^2] = h$.
- ▶ It improves pathwise local accuracy.
- ▶ In scalar problems, it typically raises the strong order from $1/2$ to 1 .

Coupling Coarse and Fine Paths

If one coarse step contains q fine steps, define

$$\Delta W_k^{\text{coarse}} = \sum_{j=1}^q \Delta W_{kq+j}^{\text{fine}}.$$

Then

$$\Delta W_k^{\text{coarse}} \sim \mathcal{N}(0, qh_{\text{fine}}).$$

Coupled comparison

Coarse and fine solutions use the same Brownian path; their difference primarily measures discretisation error.

Independent comparison

Separate random inputs mainly measure the difference between two random paths.

Strong and Weak Errors

Strong error: are paths close?

$$e_{\text{strong}}(h) = \mathbb{E}|Z_T^h - Z_T|.$$

Both values must share the same Brownian path.

Weak error: are statistics close?

$$e_{\text{weak}}(h) = |\mathbb{E}\varphi(Z_T^h) - \mathbb{E}\varphi(Z_T)|.$$

Only the expectation of a chosen observable is compared.

They are not interchangeable

A method may be inaccurate path by path while estimating a particular expectation accurately.

Reading a Convergence Order

If $e(h) \approx Ch^p$, then

$$\log e \approx \log C + p \log h.$$

Hence a log-log slope estimates the order p .

Order	factor when h is halved	two successive halvings
$p = 1/2$	$2^{-1/2} \approx 0.707$	error approximately halves
$p = 1$	$2^{-1} = 0.5$	error is approximately quartered

A slope must be interpreted together with its fitting range, uncertainty, and reference-solution error.

2. Model 1: Verification against an Exact Solution

Geometric Brownian Motion as a Test Equation

$$\boxed{dS_t = \mu S_t dt + \sigma S_t dW_t}, \quad S_0 = 100, \mu = 0.05, \sigma = 0.30, T = 1.$$

- ▶ Drift and diffusion are proportional to the current state.
- ▶ Financially, it is a simple price model; numerically, it is a solvable test equation.
- ▶ Its exact solution permits direct error measurement.
- ▶ The exact state remains positive, providing an additional diagnostic.

Why begin with this model?

A solver should reproduce known benchmark behaviour before it is used on a model without an exact solution.

Deriving the Exact GBM Solution

Apply Itô's formula to $f(S) = \log S$:

$$d \log S_t = \frac{1}{S_t} dS_t - \frac{1}{2} \frac{1}{S_t^2} (\sigma S_t)^2 dt = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t.$$

Therefore

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right].$$

Interpretation

The term $-\sigma^2/2$ follows from the stochastic chain rule; it is not an additional financial assumption.

Exact Moments as Distribution-Level Checks

$$\mathbb{E}[S_T] = S_0 e^{\mu T}, \quad \text{Var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1).$$

For the reported parameters,

$$\mathbb{E}[S_T] = 105.1271, \quad \text{Var}(S_T) = 1040.7868.$$

Exact terminal sampling produced

$$\widehat{\mathbb{E}}[S_T] = 105.1987, \quad \widehat{\text{Var}}(S_T) = 1045.0120.$$

These differences are consistent with Monte Carlo sampling variation.

Three One-Step Updates for GBM

Let $\Delta W_n \sim \mathcal{N}(0, h)$.

$$\text{Exact: } S_{n+1} = S_n \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) h + \sigma \Delta W_n\right],$$

$$\text{Euler-Maruyama: } S_{n+1} = S_n (1 + \mu h + \sigma \Delta W_n),$$

$$\text{Milstein: } S_{n+1} = S_n \left[1 + \mu h + \sigma \Delta W_n + \frac{1}{2}\sigma^2 ((\Delta W_n)^2 - h)\right].$$

Fair comparison

All three updates reuse the same increments, so terminal differences reflect numerical error rather than different random inputs.

Core GBM Implementation

```
h = T / N
dW = np.sqrt(h) * rng.standard_normal((M, N))

S_em = np.full(M, S0)
S_mil = np.full(M, S0)
for n in range(N):
    dw = dW[:, n]
    S_em *= 1.0 + mu*h + sigma*dw
    S_mil *= (1.0 + mu*h + sigma*dw
              + 0.5*sigma**2*(dw**2 - h))

W_T = dW.sum(axis=1)
S_exact = S0*np.exp((mu - 0.5*sigma**2)*T + sigma*W_T)
```

- ▶ The arrays advance M paths simultaneously.
- ▶ The exact value uses the sum of the same increments.

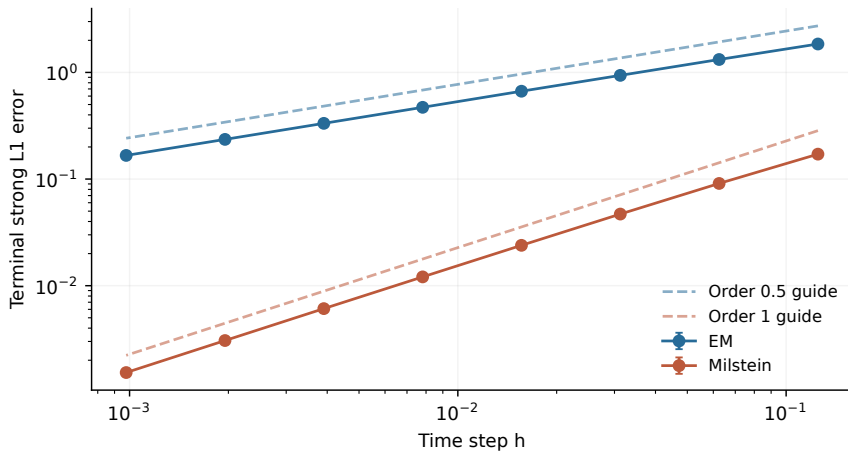
GBM Experimental Protocol

- ▶ $T = 1$, grid sizes $N = 2^k$, and time step $h = T/N$.
- ▶ A fixed number of paths and a recorded random seed are used.
- ▶ Strong error: $\hat{e} = M^{-1} \sum_i |S_{T,i}^h - S_{T,i}^{\text{exact}}|$.
- ▶ Its standard error is computed from the pathwise absolute differences.
- ▶ Weak mean error is evaluated analytically whenever possible.
- ▶ Log-log regression is applied over a stated asymptotic range.

Role of the seed

A fixed seed does not improve accuracy. It permits exact reproduction of the sample.

GBM Strong-Convergence Results



Fitted slopes: EM 0.500; Milstein 0.991.

Interpreting the Strong-Error Plot

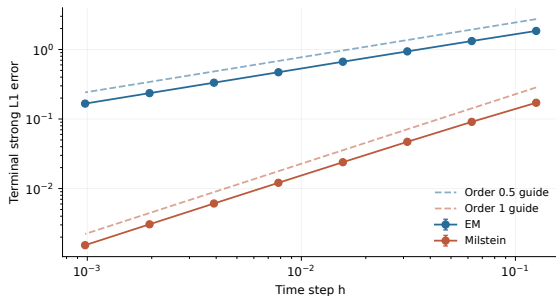
Method	$N = 64$	$N = 1024$
EM	0.66641	0.16658
Milstein	0.02391	0.00153

- ▶ Increasing N by 16 reduces h by 16.
- ▶ EM error decreases by approximately $4 = 16^{1/2}$.
- ▶ Milstein error decreases by approximately $15.6 \approx 16^1$.

Conclusion

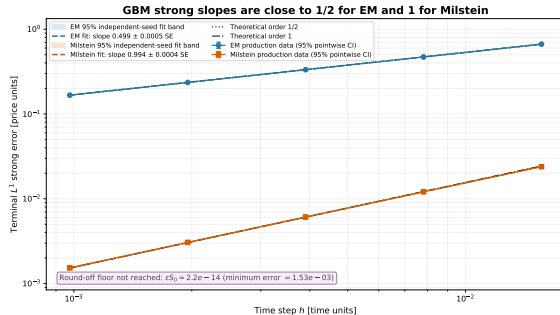
The data agree with strong order $1/2$ for EM and strong order 1 for Milstein.

Individual Challenge: Before and After



Before: error curves alone do not quantify slope stability.

The revision turns a qualitative visual claim into quantitative, reproducible evidence.



After: repeated-run error bars, theory guides, and fitted slopes.

Analytic Weak Mean Error

Euler–Maruyama satisfies

$$\mathbb{E}[S_{n+1}^{\text{EM}} \mid S_n^{\text{EM}}] = S_n^{\text{EM}}(1 + \mu h),$$

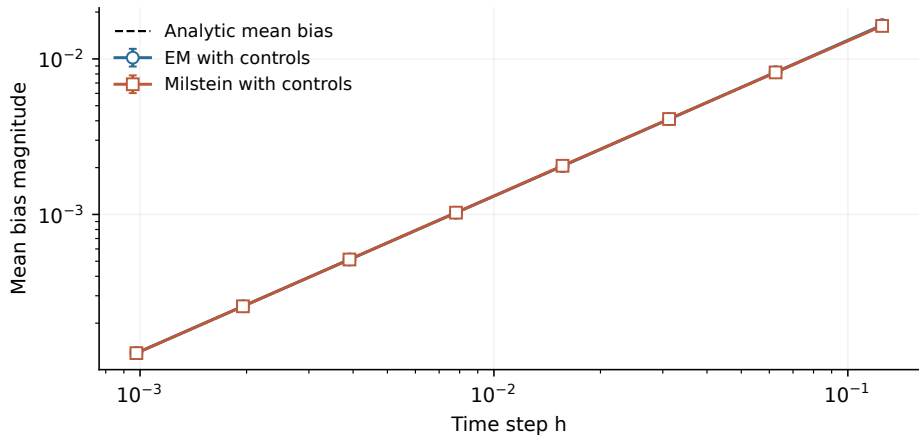
and hence

$$\mathbb{E}[S_N^{\text{EM}}] = S_0(1 + \mu h)^N, \quad N = T/h.$$

The Milstein correction has zero expectation, so both methods have the same mean. Comparing with $S_0 e^{\mu T}$ gives

$$e_{\text{weak}}(h) = \left| S_0(1 + \mu h)^{T/h} - S_0 e^{\mu T} \right| = O(h).$$

GBM Weak-Convergence Results



EM slope 0.999 and Milstein slope 1.000. The curves coincide because both methods have the same analytic mean bias.

Why Better Paths Do Not Improve This Mean

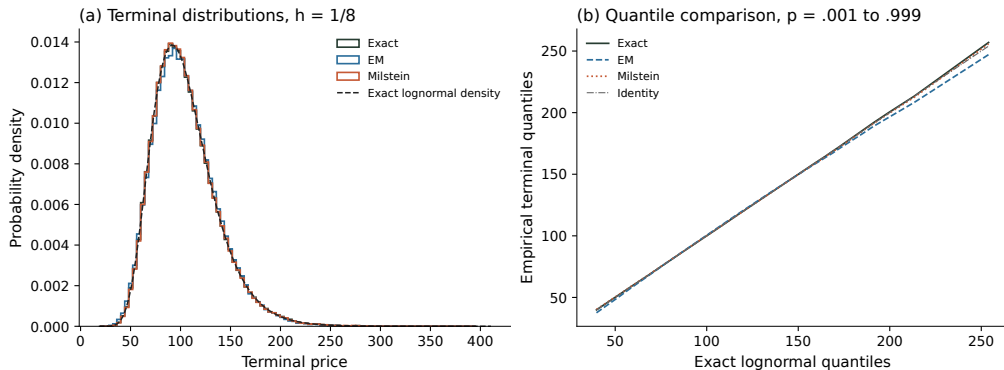
- ▶ Milstein improves the agreement of each numerical path with the exact path.
- ▶ Its additional correction has zero mean.
- ▶ For $\varphi(S) = S$, EM and Milstein therefore have identical weak errors.
- ▶ A different nonlinear observable may produce a different comparison.

Orders must be associated with an error criterion

Milstein has higher *strong* order in this benchmark. This does not imply higher weak order for every observable.

Distribution Check

Distribution sanity check | all simulated endpoints included



The histogram should agree with the exact lognormal density, while sample moments should agree within plausible Monte Carlo uncertainty.

Positivity Check

The exact solution is strictly positive. A discrete method need not preserve this property.

Terminal-value method	negative values / paths	empirical frequency
Exact update	0/160000	0.000%
Euler–Maruyama	0/160000	0.000%
Milstein	0/160000	0.000%

Scope

This is an empirical terminal-time frequency for the reported parameters and grids. It is not a proof of positivity preservation for all cases.

What the GBM Benchmark Established

- ① Brownian increments use the correct $\sqrt{h}$ scaling.
- ② Exact and numerical values are coupled through the same path.
- ③ EM and Milstein reproduce the expected strong orders.
- ④ Both methods have first-order weak mean error for a precise analytic reason.
- ⑤ Moment, distribution, and positivity checks support the implementation.

Next step

Replace the unavailable exact solution by a controlled fine-grid numerical reference.

3. Model 2: Nested-Grid Verification without an Exact Solution

A Non-Affine Model with State-Dependent Volatility

Let $S_t = e^{X_t}$ and consider

$$dX_t = \left(\mu - \frac{1}{2}g(Y_t)^2 \right) dt + g(Y_t) dW_t^{(1)},$$

$$dY_t = \kappa(\theta - Y_t) dt + \xi \sqrt{1 + Y_t^2} dW_t^{(2)},$$

$$g(y) = 0.1 + \frac{0.4}{1 + e^{-y}}, \quad \text{Corr}(dW^{(1)}, dW^{(2)}) = \rho.$$

- ▶ Y_t changes the noise amplitude of X_t .
- ▶ The coefficient functions are non-affine.
- ▶ No directly usable exact terminal solution is available.

Parameter Set

Parameter	Numerical role	Value
S_0, Y_0	initial conditions	100, 0
μ	mean growth component of X	0.05
κ, θ	mean-reversion rate and centre for Y	2, -0.2
ξ	noise scale in Y	0.6
ρ	correlation between random drivers	-0.7
T	terminal time	1

Minimal financial interpretation

Treat S as a positive state and Y as a second state controlling its instantaneous random fluctuation. No pricing theory is required.

Generating Correlated Brownian Increments

Generate independent $Z_1, Z_2 \sim \mathcal{N}(0, 1)$ and define

$$\Delta W_1 = \sqrt{h}Z_1, \quad \Delta W_2 = \sqrt{h} \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right).$$

Then

$$\text{Var}(\Delta W_1) = \text{Var}(\Delta W_2) = h, \quad \text{Corr}(\Delta W_1, \Delta W_2) = \rho.$$

Required diagnostic

The observed sample correlation was -0.699675 , compared with the target -0.7 .

Why Discretise $X = \log S$?

A direct update of S may become negative after a sufficiently large negative random increment.
Updating X and transforming back gives

$$S_n = e^{X_n} > 0$$

whenever X_n remains finite.

A structure-aware numerical choice

The variable transformation uses analytic structure to preserve a state constraint, in the same spirit as structure-aware PDE discretisations.

Synchronous Euler–Maruyama Update

Both components use the old state Y_n :

$$X_{n+1} = X_n + \left(\mu - \frac{1}{2}g(Y_n)^2 \right) h + g(Y_n)\Delta W_{1,n},$$

$$Y_{n+1} = Y_n + \kappa(\theta - Y_n)h + \xi\sqrt{1 + Y_n^2}\Delta W_{2,n},$$

$$S_{n+1} = e^{X_{n+1}}.$$

Meaning of synchronous

X_{n+1} and Y_{n+1} must both use Y_n . Updating Y first and using the new value in the X step would implement a different scheme.

Implementation Details

```
def g(y):  
    # stable sigmoid avoids exponential overflow  
    return 0.1 + 0.4 * expit(y)  
  
for n in range(N):  
    y_old = y.copy()  
    gy = g(y_old)  
    x += (mu - 0.5*gy**2)*h + gy*dW1[:, n]  
    y += kappa*(theta - y_old)*h \  
        + xi*np.sqrt(1.0 + y_old**2)*dW2[:, n]  
  
assert np.isfinite(x).all() and np.isfinite(y).all()  
s = np.exp(x)
```

Recorded diagnostics include finite-state checks, increment variances, correlation, and the number of non-positive prices.

An Exact Structural Property of the Discrete Mean

Conditional on $\mathcal{F}_n$, $g(Y_n)$ is known. Since

$$\mathbb{E}[e^{a\Delta W}] = e^{a^2 h/2},$$

$$\begin{aligned}\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] &= S_n e^{(\mu - \frac{1}{2}g_n^2)h} \mathbb{E}[e^{g_n \Delta W_{1,n}}] \\ &= S_n e^{\mu h}.\end{aligned}$$

Hence

$$\mathbb{E}[S_N] = S_0 e^{\mu T} \quad \text{for every step size } h.$$

Consequence for weak testing

The observable $\varphi(S) = S$ has zero time-discretisation bias and cannot test whether the terminal distribution is approximated correctly.

Why Use a Call Payoff as the Nontrivial Observable?

Choose

$$\varphi(S) = (S - K)^+ = \max(S - K, 0), \quad K = 100.$$

- ▶ It is nonlinear, so $\mathbb{E}[\varphi(S)]$ is not determined by $\mathbb{E}[S]$.
- ▶ Equal means do not imply equal payoff expectations.
- ▶ It detects changes in the shape and upper tail of the terminal distribution.
- ▶ Its kink at $S = K$ makes the weak rate more difficult to resolve numerically.

Reason for the choice

Once the linear mean is exact by construction, a distribution-sensitive observable is required. Detailed financial interpretation is unnecessary.

Nested Fine-Grid Reference

Use the same finest Brownian increments to construct

$$S_T^N \quad \text{and} \quad S_T^{N_{\text{ref}}}, \quad N_{\text{ref}} \gg N.$$

The coupled strong difference is

$$\widehat{D}_N = \frac{1}{M} \sum_{i=1}^M \left| S_{T,i}^N - S_{T,i}^{N_{\text{ref}}} \right|.$$

For a weak observable, use the paired difference

$$\widehat{B}_N = \frac{1}{M} \sum_{i=1}^M \left[\varphi(S_{T,i}^N) - \varphi(S_{T,i}^{N_{\text{ref}}}) \right].$$

Precise terminology

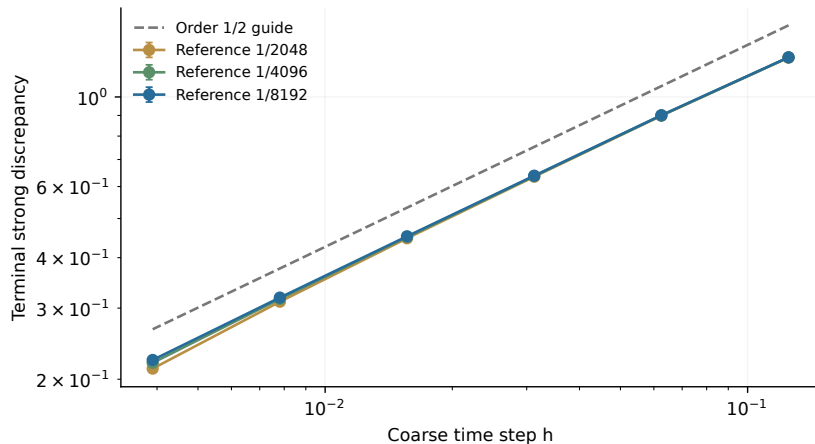
These are differences relative to a numerical reference, not known exact errors. Reference-grid sensitivity must therefore be investigated.

Nested-Grid Experimental Design

Component	Setting
Coarse grids	$N = 8, 16, 32, 64, 128, 256$
Main reference grid	$N_{\text{ref}} = 8192$
Sensitivity references	2048, 4096, 8192
Number of paths	$M = 100000$
Coupling	all grids aggregate the same finest increments
Strong quantity	$\mathbb{E} S_T^N - S_T^{N_{\text{ref}}} $
Weak quantities	mean difference and paired call-payoff difference

The strong-rate fit uses $N = 8, 16, 32, 64$ to limit curvature caused by the finite reference grid at the finest coarse levels.

Non-Affine Strong-Difference Results



The fitted slope relative to $N_{\text{ref}} = 8192$ is 0.491, consistent with the typical strong order 1/2 of Euler–Maruyama.

Reference-Grid Sensitivity

Hold the fitted coarse grids $N = 8, 16, 32, 64$ fixed:

N_{ref}	fitted slope	interpretation
2048	0.496	coarser reference, similar slope
4096	0.493	close to the finer result
8192	0.491	main reported reference

Supported conclusion

The approximately half-order slope is stable under these reference choices. Their agreement is numerical evidence, not an exact error bound.

The Reference Also Has Discretisation Error

Differences between successive reference levels are

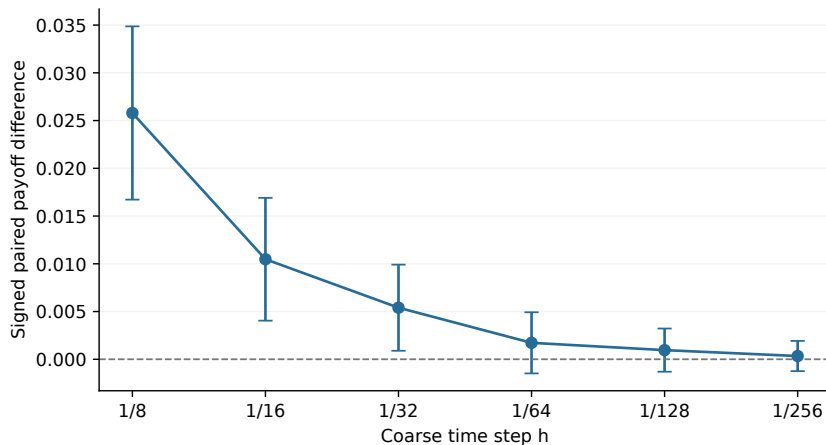
$$\mathbb{E}|S_T^{2048} - S_T^{4096}| \approx 0.05679, \quad \mathbb{E}|S_T^{4096} - S_T^{8192}| \approx 0.04016.$$

Their ratio is

$$\frac{0.04016}{0.05679} \approx 0.707 \approx 2^{-1/2}.$$

- ▶ The reference levels themselves show half-order reduction.
- ▶ The 8192-step solution is not exact.
- ▶ It is sufficiently fine to reveal a stable slope for $N = 8$ to 64 .

Non-Affine Weak Differences



Coarse-grid payoff differences are statistically resolved. At finer grids, confidence intervals include zero and sampling noise is comparable with the discretisation difference.

Why No Reliable Weak Order Is Reported

- ▶ For $N = 8, 16, 32$, paired-payoff confidence intervals exclude zero.
- ▶ For $N = 64, 128, 256$, the intervals include zero.
- ▶ At fine levels, the sign and magnitude are sensitive to sampling noise.
- ▶ A log-log fit would mistake noise for an asymptotic convergence law.
- ▶ The payoff kink also makes the asymptotic regime harder to resolve.

Correct interpretation

Weak differences decrease, but the available sample budget is insufficient for a credible weak-order estimate. This limits the evidence; it does not demonstrate failure of the numerical scheme.

Sample-Size Experiment

Fix $N = 64$ and $N_{\text{ref}} = 8192$:

M	paired difference	standard error	95% interval
10,000	0.009045	0.005149	$[-0.001048, 0.019137]$
40,000	0.004942	0.002592	$[-0.000139, 0.010022]$
100,000	0.001722	0.001633	$[-0.001479, 0.004923]$

- ▶ The standard error decreases approximately as $M^{-1/2}$.
- ▶ All three intervals still include zero.
- ▶ A larger budget or variance reduction is needed for rate estimation.

Mean and Implementation Diagnostics

The exact discrete-mean identity predicts

$$\mathbb{E}[S_T^N] = 100e^{0.05} = 105.1271.$$

The finest simulation produced

$$\hat{\mathbb{E}}[S_T] = 105.08796, \quad 95\% \text{ CI } [104.89962, 105.27629],$$

which contains the theoretical value. The associated payoff estimate was

$$14.53804, \quad 95\% \text{ CI } [14.40515, 14.67093].$$

Different roles

The mean checks the implementation; the nonlinear payoff probes distributional weak error.

Numerical Diagnostic Checklist

Diagnostic	Observation
increment variances	consistent with $1/8192 \approx 1.221 \times 10^{-4}$
increment correlation	-0.699675 , target -0.7
finite states	all reported X , Y , and S values were finite
positivity	$S = e^X$ remained positive
exact mean identity	theoretical mean lies inside the empirical interval
reference sensitivity	three reference grids give similar strong slopes

A visually convincing convergence plot does not replace these checks.

4. Evaluation, Reproducibility, and Conclusions

What the Coursework Requires

Component	Required task	Evidence reported here
GBM strong convergence	compare EM and Milstein with theory	slopes 0.500 and 0.991
GBM weak behaviour	choose and explain an observable	analytic first-order mean bias
distribution / positivity	quantitative checks	moments, histogram, negative frequency
non-affine study	nested grids and sensitivity	stable half-order trend
non-affine study	intervals and justified conclusion	trend visible; rate unresolved
reproducibility	commands, seed, environment, Git	traceable code and outputs

A weak order need not be forced from unresolved data. The requirement is a sound experiment and an evidence-based interpretation.

Code Structure Mirrors the Mathematics

- ① **Parameters:** initial data, coefficients, T , grids, paths, seed.
- ② **Random increments:** finest-grid normal samples and correlation transform.
- ③ **Solvers:** EM, Milstein, and log-variable EM.
- ④ **Coupling:** aggregation of fine increments for each coarse grid.
- ⑤ **Statistics:** means, standard errors, intervals, fitted slopes.
- ⑥ **Outputs:** tables, figures, data, and run metadata.

Practical benefit

An anomalous plot can be traced through statistic, path, increment, and parameter layers.

A Minimal Reproducibility Chain

Identify the Git commit containing code, parameters, and report



Run the exact recorded command line



Fix the seed and record Python / NumPy versions



Save raw output and figure data



Match the reported number to the required precision

If a script and report disagree, correct the report or identify the version difference.

Eight Common Failure Modes

- ❶ using hZ instead of $\sqrt{h}Z$ for Brownian increments;
- ❷ comparing independent paths when measuring strong error;
- ❸ confusing sample standard deviation with standard error;
- ❹ refining h without checking the Monte Carlo noise floor;
- ❺ updating Y before using it in the synchronous X update;
- ❻ describing a finite fine-grid solution as exact;
- ❼ fitting a weak rate to sign-unstable differences whose intervals include zero;
- ❽ reporting a slope without the fitting range, uncertainty, and diagnostics.

How to Read a Convergence Plot Critically

Ask the following questions in order:

- 1 Is the horizontal axis h , N , or computational cost?
- 2 Is the vertical quantity an exact error or a reference difference?
- 3 Are coarse and fine solutions driven by the same random input?
- 4 Do error bars show standard deviation, standard error, or confidence intervals?
- 5 Which points are used in the slope fit, and why?
- 6 Do the finest points reach a noise or reference-error floor?
- 7 Is the conclusion stable under changes in sample size or reference grid?

Practical rule

Identify what the vertical axis measures before assessing linearity or reading a slope.

Main Conclusions

- ① EM and Milstein show GBM strong orders close to $1/2$ and 1 .
- ② For the terminal mean, both methods share the same first-order weak bias.
- ③ Nested-grid experiments support half-order strong convergence in the non-affine model.
- ④ The non-affine discretisation preserves the mean exactly, so a nonlinear payoff is needed to detect distributional weak error.
- ⑤ The present path budget supports a decreasing weak-difference trend, but not a credible weak-order estimate.
- ⑥ Confidence intervals, reference sensitivity, and diagnostics are as important as slopes.

Possible Extensions

- ① Increase paired samples or use a control variate for payoff differences.
- ② Balance h and M under a fixed computational budget.
- ③ Refine the reference grid and quantify its systematic contribution.
- ④ Report uncertainty in the fitted convergence slope.
- ⑤ Compare Monte Carlo with a finite-difference solution of the associated backward PDE for a simplified one-dimensional model.

Scope

These are natural extensions. The present coursework should first explain its existing evidence and limitations clearly.

Thank you

Questions?

Starting from the Euler method, three additions were essential: **normal increments, repeated sampling, and quantified uncertainty.**

5. Backup Derivations

Backup: Why Is the GBM Mean Bias First Order?

Let $N = T/h$. Since

$$\log(1 + \mu h) = \mu h - \frac{1}{2}\mu^2 h^2 + O(h^3),$$

$$\begin{aligned}(1 + \mu h)^{T/h} &= \exp\left[\frac{T}{h} \log(1 + \mu h)\right] \\ &= \exp\left[\mu T - \frac{1}{2}\mu^2 Th + O(h^2)\right] \\ &= e^{\mu T} \left(1 - \frac{1}{2}\mu^2 Th + O(h^2)\right).\end{aligned}$$

Therefore

$$S_0(1 + \mu h)^{T/h} - S_0 e^{\mu T} = O(h).$$

Backup: Why Pairing Reduces Noise

Compare

independent: $\widehat{\mathbb{E}}[\varphi(S^N)] - \widehat{\mathbb{E}}[\varphi(S^{\text{ref}})],$

paired: $\frac{1}{M} \sum_i [\varphi(S_i^N) - \varphi(S_i^{\text{ref}})] .$

For a paired difference,

$$\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B).$$

A shared Brownian path makes A and B strongly positively correlated, so common path fluctuations cancel in the difference.

Backup: Standard Error and Sample Size

For paired samples $D_1, \dots, D_M$,

$$\bar{D} = \frac{1}{M} \sum_i D_i, \quad s_D^2 = \frac{1}{M-1} \sum_i (D_i - \bar{D})^2, \quad \text{SE}(\bar{D}) = \frac{s_D}{\sqrt{M}}.$$

To reduce the standard error from ε_1 to ε_2 ,

$$M_2 \approx M_1 \left(\frac{\varepsilon_1}{\varepsilon_2} \right)^2.$$

Reducing it by a factor of five requires approximately twenty-five times as many paths.

Backup: Terminology

Term		Meaning in this coursework
drift		deterministic mean change per unit time
diffusion		amplitude of the random fluctuation
Brownian increment	increment	normal variable with mean 0 and variance h
path		a complete sequence $Z_0, Z_1, \dots, Z_N$
observable		a terminal function $\varphi(Z_T)$
strong convergence	convergence	numerical and exact values approach on the same path
weak convergence		expectations of an observable approach
nested grid		coarse increments are sums of shared fine increments
numerical reference	reference	a fine-grid approximation used for an unknown exact solution

References and Project Files



Course Problem Pack, Direction 4: Financial SDEs, pp. 6–7.



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SIAM Review, 43(3), 2001.



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Numerical Solution of Stochastic Differential Equations,
Springer, 1992.

Project scripts: `gbm.py`, `nonaffine.py`; undergraduate report: `undergraduate_revision/`.